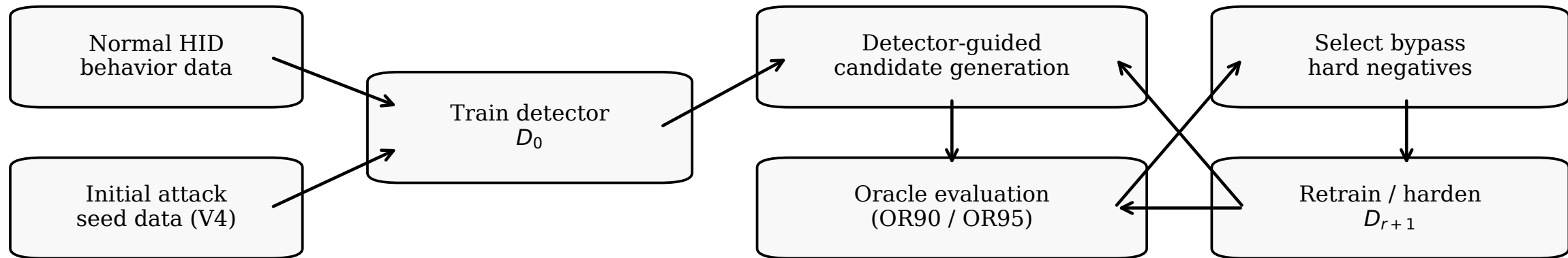


Detector-guided hard-negative hardening framework



Each round searches for detector blind spots, promotes successful bypass cases to hard negatives, and retrains the next detector using the augmented training set.